

# DS 442 Project 1 Report

## Question 1.2 Part 1

- *The solution of Q1.1.a (DFS) is:*

Solution Path: (3, 3, 0, 0, L) -> (3, 1, 0, 2, R) -> (3, 2, 0, 1, L) -> (3, 0, 0, 3, R) -> (3, 1, 0, 2, L) -> (1, 1, 2, 2, R) -> (2, 2, 1, 1, L) -> (0, 2, 3, 1, R) -> (0, 3, 3, 0, L) -> (0, 1, 3, 2, R) -> (0, 2, 3, 1, L) -> (0, 0, 3, 3, R)

Total cost = 11

Number of node expansions = 12

- *The solution of Q1.1.b (BFS) is:*

Solution Path: (3, 3, 0, 0, L) -> (3, 1, 0, 2, R) -> (3, 2, 0, 1, L) -> (3, 0, 0, 3, R) -> (3, 1, 0, 2, L) -> (1, 1, 2, 2, R) -> (2, 2, 1, 1, L) -> (0, 2, 3, 1, R) -> (0, 3, 3, 0, L) -> (0, 1, 3, 2, R) -> (0, 2, 3, 1, L) -> (0, 0, 3, 3, R)

Total cost = 11

Number of node expansions = 14

## Question 1.2 Part 2

Both DFS and BFS found the same solution of 11 trips, but DFS expanded 12 nodes while BFS expanded 14. BFS guarantees the optimal solution because it checks every state one trip away before any state two trips away (BFS checks adjacent branches before deeper exploration), so the goal it reaches uses the fewest trips, whereas DFS on the other hand fully explores one branch and returns the first goal it finds, so it may end up missing a shorter path. DFS expanded 12 states while BFS expanded 14, so DFS did slightly less work for the same answer, this tells us that BFS spent extra expansions to guarantee the shortest path.